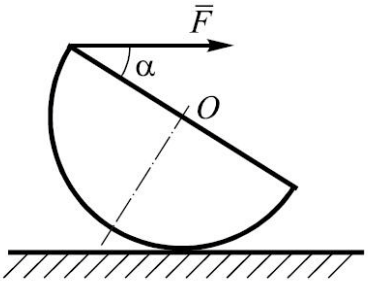
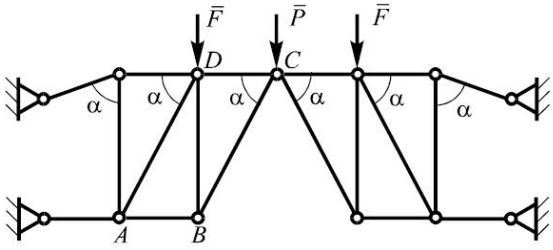
Problem S2-2009



The semicylinder in the scheme contacts with the rough surface and is loaded with a certain horizontal force *F*. The coefficient of friction between the semicylinder and the surface is *f*. The cylinder density is linearly varying according to the distance from the semicylinder selected point to the centre *O*. And the curvilinear surface material density is twice as high as the density at *O* point. Find the maximal angle α when the system is balanced.

图示半圆柱与粗糙的水平面接触，受到一水平力 F 作用。半圆柱与水平面之间的摩擦系数为 f；半圆柱的密度随半圆柱上的点到中心点 O 的连线距离而呈线性变化，且半圆柱表面曲面的材料密度是 O 点密度的两倍。求系统平衡时的最大角度 α。

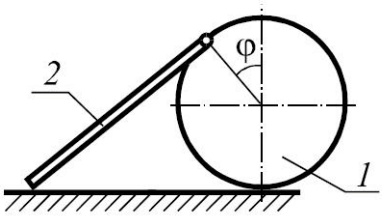
Problem S1 – 2010 (8 points)



In the depicted rod construction all rods are weightless. Find at what proportion of the internal forces F and P in the rods AB and CD are 5 times different. $\alpha = 60°$.

如图所示杆件系统，所有杆件自重均不计。求力 ***F*** 和 ***P*** 是什么比例时，杆 ***AB*** 的内力与 ***CD*** 内力相差 5 倍？α=60°。

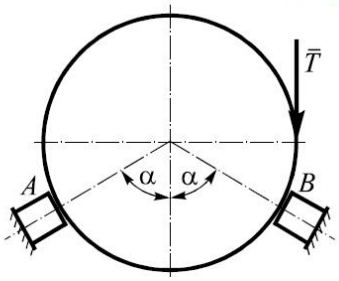
Problem S2 – 2010 (7 points)



The homogeneous disc with the mass m_1 and the radius R is hinged to the homogeneous rod with the mass m_2 and the length l. Define the minimal constant of friction between the bodies and the surface when the system is in equilibrium in the position defined by the angle φ.

半径为 R、质量为 m_1 的均质圆盘与质量为 m_2、长为 l 的均质杆铰接。若系统在图示角φ处平衡，试求圆盘和杆与支持面之间的摩擦系数的最小值。

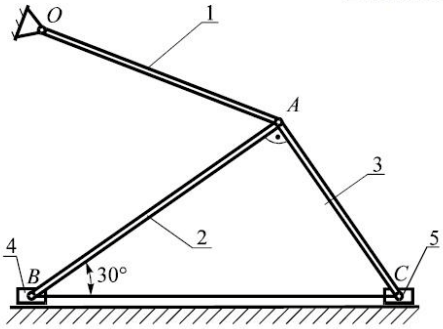
Problem S1-2011 (10 points)



The homogeneous cylinder with weight G lies on two symmetrical to vertical supports A and B, the positions of which are defined by angles α. Coefficient of friction between the cylinder and supports is f. Find the minimum force value T to be applied to disturb the cylinder from equilibrium.

重为 G 的均质圆柱放置在两个对称于垂线的支座 A 和 B 上，支座 A 和 B 的位置由角度α确定，如图所示。圆柱和支撑之间的摩擦系数为 f 。求打破圆柱平衡的最小力 T 的值。

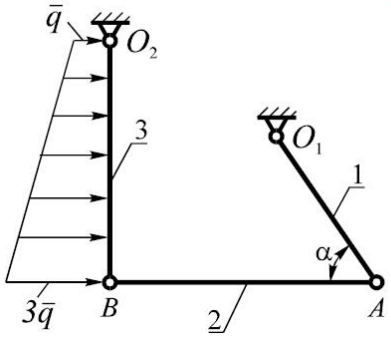
Problem S2-2011 (10 points)



Construction lies in the vertical plane. It consists of three homogeneous rods 1, 2 and 3of the same masses which connected by hinge at point A and weightless sliders 4 and 5 tied together by the inextensible cord. Define the ratio between the largest of the three forces of rods acting on hinge *A* and the smallest analogous force.

位于铅垂面内的结构，由铰接在 A 点的 1、2 和 3 三个同样质量的均质杆及不计自重的滑块 4、5 组成，4、5 之间由不可伸长的细绳连接。求三根杆在 A 点受到的力最大值和最小值的比。

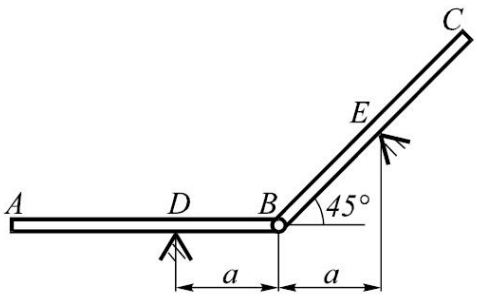
Problem S1-2012



In the bar system placed in the vertical plane the homogeneous rods 1, 2 and 3 have lengths l_1, l_2, l_3 and gravity forces G_1, G_2, G_3 respectively. In equilibrium position rod AB is horizontal, rod O_2B is vertical and the angle between rods O_1A and AB is α. Define the value q for the intensity of the distributed load with linear variation when the given equilibrium position of system is provided.

位于铅垂面内的杆件系统中，均质杆 1、2 和 3 长度分别为 l_1、l_2、l_3，重分别为 G_1 、G_2，、G_3。平衡时，杆 AB 水平，杆 O_2B 垂直，杆 O_1A 与 AB 间的夹角为α。求给定系统在图示位置平衡时，线性分布载荷的集度 q 的值。

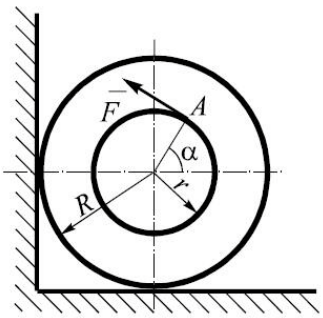
Problem S2-2012



Two homogeneous rods AB and BC with the same masses are connected by hinge at point B and placed on two rough edges D and E. Define the rod lengths when the equilibrium in the depicted position is possible if the coefficient of friction between the rods and supports is f, and size a is also known.

两根相同质量的均质杆 AB 和 BC 铰接在 B 点并放在两个粗糙的棱角 D 和 E 上。若两杆与支撑之间的摩擦系数为 f，尺寸 a 已知，求系统在图示位置平衡时，两杆的长度应该为多少？

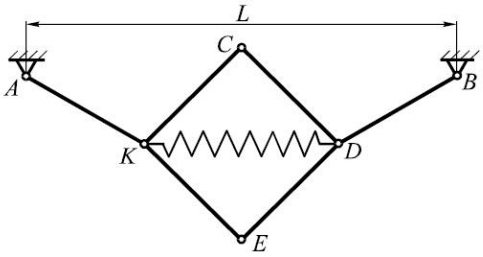
Problem S1–2013



Step block has weight G and lies on a smooth horizontal surface. The block is in contact with rough vertical wall. Radii R and r are known. Coefficient of static friction of the block with the wall is equal to f. Force F is applied tangentially to a circle at point A which position is determined by the angle α. For what values of force F, the block will be in equilibrium?

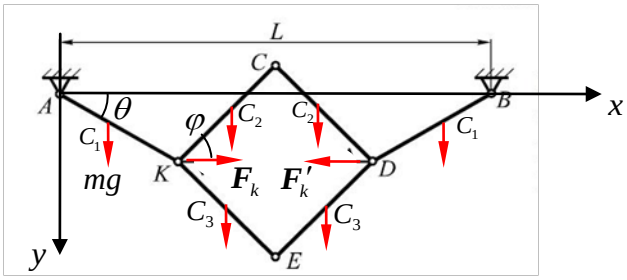
塔轮重 G，放在光滑的水平面上，并与粗糙的墙壁接触，半径 R 和 r 已知。塔轮与墙壁间的静摩擦系数为 f。力 F 作用在点 A 处切线方向，确定 A 点位置的角 α 已知。力 F 的值为多少时，塔轮将处于平衡？

Problem S2–2013



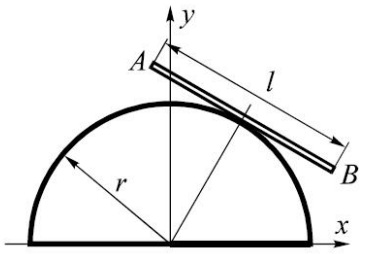
The system of related identical homogeneous rods which have length ***l*** and mass ***m***, is located in a vertical plane. Rods ***KC***, ***CD***, ***DE***, and ***EK*** form a square in the position of equilibrium. Determine the force of the spring ***KD***, if $L = l(\sqrt{2}+\sqrt{3})$.

由相同的长为 ***l***、为 ***m*** 的均质杆组成的系统位于垂直面内。杆 ***KC***、***CD***、***DE*** 和 ***EK*** 形成一个平衡状态的正方形。若 $L = l(\sqrt{2}+\sqrt{3})$，求弹簧 ***KD*** 的力。



Problem S1-2014

Problem S1–2014

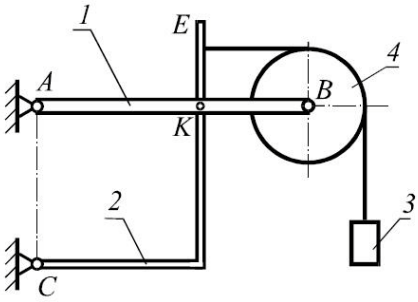


Homogeneous thin rod AB of length l is based on the half-cylinder of radius r. Static friction coefficient between the rod and the half cylinder is equal to f, the coefficient of rolling friction is δ.

Find the maximal possible value of the x coordinate of the rod point B for the case of rod's equilibrium.

均质细杆 AB，长为 l，放在半径为 r 的半圆柱上，杆和半圆柱间的静滑动摩擦系数为 f，滚动摩阻为δ，求杆处于平衡时的杆上 B 点的 x 坐标的最大值。

Problem S2–2014

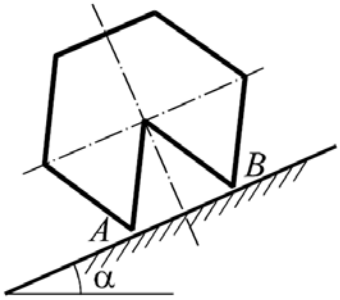


The construction shown in the figure consists of two weightless rods 1 and 2, which are connected at point K by a hinge. The load 3 is linked with the rod 2 by the weightless thread thrown over the block 4. The load 3 and block 4 have the same mass. Block 4 has the radius r, $KB = 2r$, $AK = 3r$.

Determine the distance AC when the reaction forces at points A and C are different in 2 times.

如图所示结构由不计自重的杆 1 和杆 2 在 K 处铰接组成，载荷 3 与杆 2 用无重绳子跨过滑轮 4 而连接，重物 3 和滑轮 4 质量相同，滑轮的半径为 r，KB=2r,AK=3r，试求当 A、C 两处约束力相差 2 倍时，距离 AC 是多少？

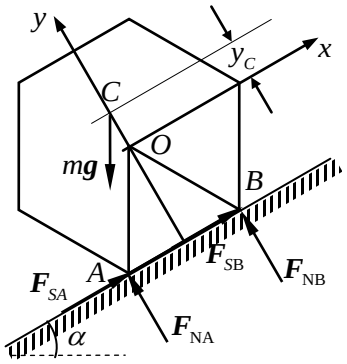
Problem C1–2015



A prism rests on an inclined plane. The base of the prism is a regular hexagon with a equilateral triangle cutout. The friction coefficient in the point A is equal to f, in the point B – $3f$. Determine the values of the plane angle α for the case of prism equilibrium.

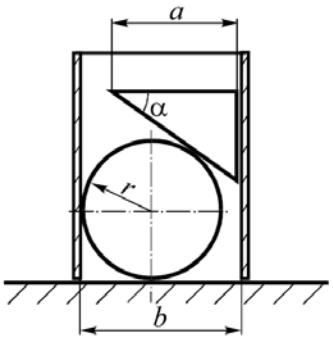
A prism rests on an inclined plane. The base of the prism is a regular hexagon with a equilateral triangle cutout. The friction coefficient in the point A is equal to f, in the point B-3f. Determine the values of the plane angle α for the case of prism equilibrium.

一棱柱体静止放置于斜面上，棱柱体的底面为一带有等边三角形切口的正六边形， ***A*** 点处摩擦系数为 ***f***，***B*** 点处摩擦系数为 ***3f***，求当棱柱体恰好处于平衡状态时，斜面的倾角 α 。



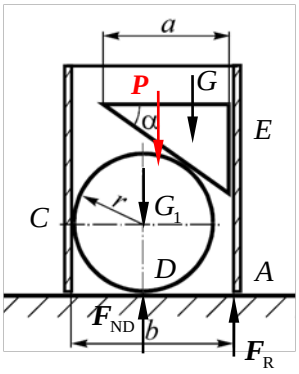
Solution

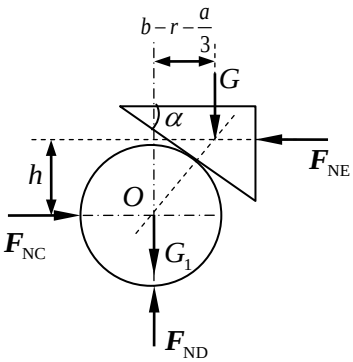
Problem C2–2015

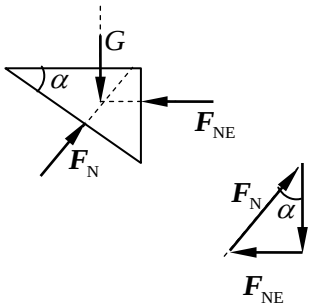


The homogeneous box of weight P and width b stands on horizontal surface. The box is symmetrical relative to a vertical plane. The homogeneous cylinder of radius r and prism with edge length a ($a < b$) are placed into the box. The prism angle α is also given. Friction forces can be neglected. Determine the values of the prism gravity force for the equilibrium of the system.

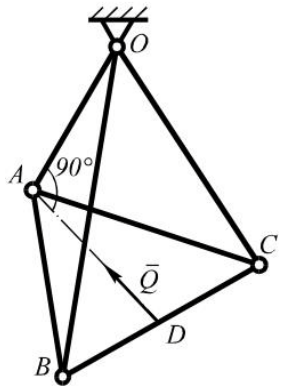
重量为 ***P***，宽度为 ***b*** 的均质盒子，直立于水平面上，盒子相对于垂直平面对称。半径为 ***r*** 的均质圆柱体和一边长为 ***a*** 的棱柱体一起放置于盒子中，棱柱体底面的 α 角已知，各处摩擦不计，求系统平衡时棱柱体的重力。







Solution1:



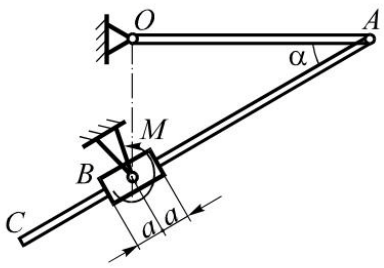
Problem S1–2016

A homogeneous plate *ABC* of weight *P*, has a form of equilateral triangle and it is hanged by three weightless rods. The rods are linked to the corners of the plate by spatial hinges. Rod *OA* is located perpendicularly to the plane of the plate. The length of rod *OA* is equal to the length of the side of the triangle *ABC*. $OB = OC$. The force *Q* is applied at point *D* and it lies in the plane of the triangle. $Q = \frac{P}{2}$.

Determine the values of reaction forces inside the rods for the case of the plate equilibrium.

Problem S1-2016 形状为等边三角形、重为 *P* 的均质板 *ABC* 用三根不计自重的杆悬挂，三杆的端部用空间铰链连接。杆 *OA* 垂直于板、长度等于板 *ABC* 的边长，*OB*=*OC*。力 *Q* 作用在点 *D* 并位于三角形平面，*Q*=*P*/2。求板 *ABC* 处于平衡时各杆的内力。

Solution. The free-body diagram of the plate ABC is drawn（shown） in the figure. There are

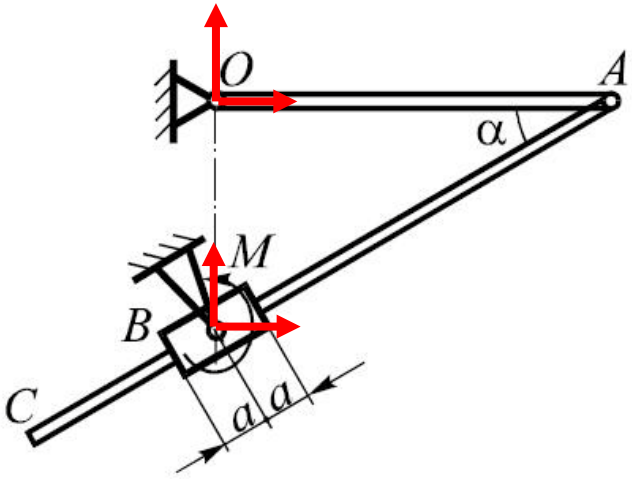


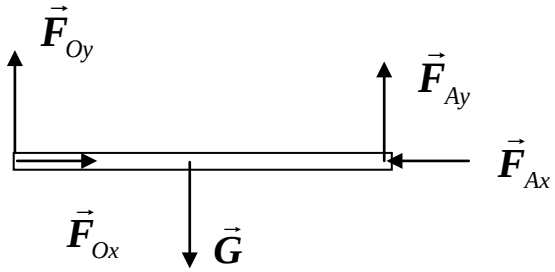
Problem S2–2016

Rods *OA* and *AC* are located in the vertical plane as shown in the figure. The weight of *OA* is *G*, the weight of *AC* is 2*G*. $OA = l, AC = 2l$. $\alpha = 30°$. The rod AC can slide relative to the rough coupling *B*, friction coefficient $f = \frac{\sqrt{3}}{20}$. $a = \frac{l}{5\sqrt{3}}$.

Find the maximal value of the balancing torque *M* applied to the coupling *B* for the case of equilibrium of the described mechanism.

Problem S2-2016 如图，杆 *OA*、*AC* 位于垂直面内，*OA* 杆重为 *G*，*AC* 杆重为 2*G*，*OA*=*l*，*AC*=2*l*，*α*=30°,杆 *AC* 相对于粗糙的滑块 *B* 滑动，摩擦系数为$f = \frac{\sqrt{3}}{20}, a = \frac{l}{5\sqrt{3}}$。求图示系统平衡时作用在滑块上的平衡力偶矩 *M* 的最大值。

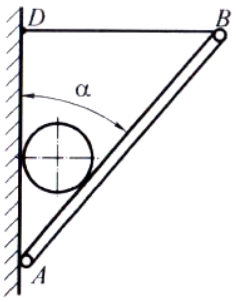




Solution:

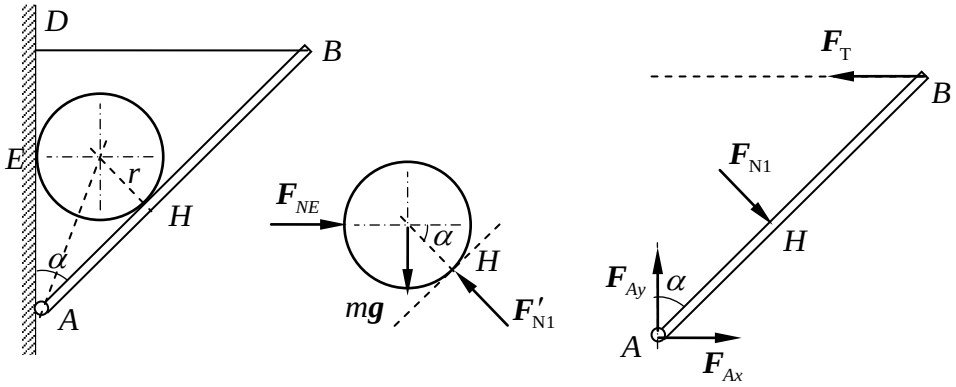
Ans.

Problem S-1



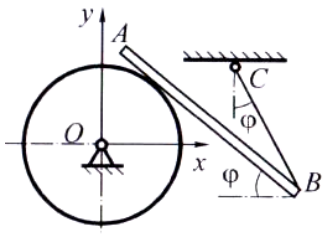
A uniform heavy cylinder touches a vertical wall and it is held in the equilibrium by two horizontal cables **BD** and two weightless rods **AB** that are pivotally attached to the wall. At what value of the angle α the tension of the cables will be the minimal?

2019　均质圆柱与铅直墙面接触并由两根水平缆绳 BD 和两根不计自重的杆 *AB* 维持平衡，杆与墙面铰接。问角*α*为多大时，缆绳的拉力为最小值？



Solution:

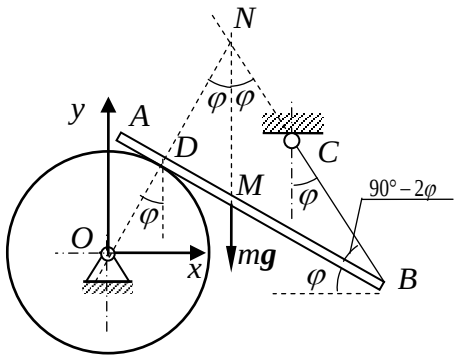
Problem S-2

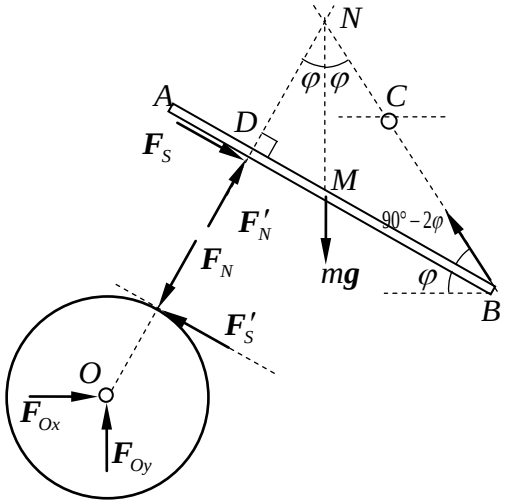


A thin homogeneous rod **AB** of length $2r$ rests on a rough disk of radius r. The rod is held in balance by a weightless thread **CB** of length r. Determine the coordinates of the point **C** that is the attachment point of the thread, if the rod is inclined to the horizontal and to the vertical at an angle φ. The friction in the hinge can be neglected.

the rod is inclined

2019 Problem S-2 长为 $2r$ 的均质细杆 AB 靠在半径为 r 的粗糙圆盘上，杆用长为 r 的不计重量的细绳 CB 维持平衡。杆倾角为φ，求细绳的悬点 C 的坐标。铰链摩擦不计。





Solution: